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INTRODUCTION

In today's fast-paced corporate and retail environments, data-driven decisions separate thriving business entities from stagnating ones. This project focuses on analysing a comprehensive corporate revenue sales dataset to uncover hidden trends, evaluate product performance, and optimize sales strategies. By leveraging a modern data stack—combining the structural flexibility of a NoSQL database with the analytical power of Power BI —this project transforms raw transactional logs into actionable business intelligence.

The primary business goals of this analysis are to solve specific operational friction points and identify key growth vectors. Specifically, the analysis is engineered to identify revenue drivers by determining which product categories and regions generate the highest profit margins. Furthermore, looking into historical data allows the business to analyse seasonality, pinpointing peak buying trends to help inventory managers optimize stock levels and prevent costly stockouts. Finally, the project handles customer segmentation, highlighting high-value customer segments to assist marketing teams in refining targeted campaigns and improving overall Company Lifetime Value.

Unlike traditional data projects that rely solely on flat CSV files or rigid relational databases, this system implements a dynamic, semi-structured data pipeline. Document-based records are securely housed within a MongoDB environment, allowing for nested customer information and variable transaction lengths. This architecture drastically reduces local memory overhead and ensures that data processing scales efficiently before advanced statistical analysis and visualization take place.

DATASET DESCRIPTION

The dataset used throughout this project contains historical transaction records for a global enterprise operating in both consumer and corporate markets. Because individual sales records inherently include nested attributes—such as detailed customer profiles and itemized arrays of purchased products—MongoDB was selected as the optimal storage engine. This document-based database format allows the dataset to scale horizontally without the rigid design constraints or intensive join operations typical of a traditional relational schema.

Each transaction is captured as a binary JSON document containing specific data fields that map out the lifecycle of a sale. Every record is uniquely identified by a default MongoDB object identifier, which sits alongside an alpha-numeric transaction sequence number and an exact timestamp detailing when the purchase occurred. The nested customer object breaks down individual buyers by their market segment, such as corporate or individual consumers, as well as their geographic region. The transactional core of the document consists of an array of items, where each element explicitly tracks the product identifier, its parent department category, the quantity of units purchased, and the unit price. Final fields record the payment method utilized and the overall fulfilment status of the order.

The structure of the data can be conceptually organized into a data dictionary that governs how the fields are treated during analysis. The raw transaction ID and order status fields function as categorical indicators, allowing for rapid filtering of valid sales revenue. Temporal trends are extracted directly from the system timestamp to map out from monthly, yearly revenue cycles. Meanwhile, numerical metrics like the quantity and unit price fields serve as the foundation for all quantitative variables, as they are multiplied dynamically during the query phase to calculate gross margins and overall corporate revenue.

IMPLEMENTATION

The implementation of this project based on heavy data aggregation tasks, which were pushed entirely to the database layer. Instead of executing a resource-heavy import of millions of raw database documents into local memory, a custom JSON aggregation pipeline was deployed directly within MongoDB. This pipeline begins by filtering out unfulfilled transactions to ensure data integrity. It then flattens the nested arrays of purchased items, groups the remaining records by their department categories, and calculates the total revenue and units sold simultaneously before sorting the final output in descending order.

By establishing a secure client connection to the database instance, the script injects the JSON aggregation pipeline payload natively into the collection environment. The aggregation pipelines were there after being analysed and pushed to JSON files.

The final stage of implementation focuses on exploratory data analysis and visual communication. Utilizing advanced visualization tools, like Microsoft Power BI, the processed data was used to generate clean visual assets, such as bar charts representing top revenue categories and line graphs illustrating seasonal revenue trends.

These visual outputs directly show the execution of the aggregation pipelines, proving that raw, unstructured transactional documents can be systematically parsed, aggregated at scale within a NoSQL database, and smoothly translated into polished business insights.

SAMPLE OUTPUTS

1. Total Revenue Generated by Company:

The analysis provides the overall performance of the company’s business from the use of Revenue Sales Dataset. It shows the total revenue generated by the company.

Total Revenue ...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

Date : "2-19-16"

Year : 2016

Month : "February"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Date : "2-20-16"

Year : 2016

Month : "February"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Stage1

\$group

1 /**

2 * _id: The id of the group.

3 * fieldN: The first field name.

4 */

5 {

6 _id: null,

7 "Revenue": {

8 \$sum: "\$ Revenue "

9 }

10 }

Output preview after \$group stage (Sample of 1 document)

_id: null

Revenue : 8244457.53

2. Profit Analysis by Category:

This aggregation analysis includes the output of profit generated by the product category section of the dataset, which works by subtracting the unit price to the unit cost of particular product.

Profit Analysis ...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Unit Cost : 3.67

Unit Price : 5

Cost : 11

Revenue : 15

Stage1

\$project

1 /**

2 * specifications: The fields to

3 * include or exclude.

4 */

5 {

6 "Product Category": 1,

7 profit: {

8 \$multiply: [

9 {

10 \$subtract: [

11 "\$ Unit Price ",

12 "\$ Unit Cost "

13]

14 },

15 "\$Quantity"

16]

17 }

18 }

Output preview after \$project stage (Sample of 10 documents)

_id: ObjectId('6a321615dc42fde44e38f409')

Product Category : "Accessories"

profit: 29

_id: ObjectId('6a321615dc42fde44e38f40c')

Product Category : "Clothing"

profit: 8

3. Total Sales Revenue by Year:

This output demonstrates the distribution of total sales in that particular year that is, 2015 or 2016.

Total Sales Res...

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 1

\$group

```

1  /**
2   * _id: The id of the group.
3   * fieldN: The first field name.
4   */
5  {
6    _id: "$Year",
7    Revenue: {
8      $sum: 1
9    }
10 }

```

Output preview after \$group stage (Sample of 3 documents)

_id: 2015

Revenue : 15019

_id: 2016

Revenue : 19847

Stage 2

\$sort

```

1  /**
2   * Provide any number of field/order pair
3   */
4  {
5    _id: -1
6  }

```

Output preview after \$sort stage (Sample of 3 documents)

_id: 2016

Revenue : 19847

_id: 2015

Revenue : 15019

4. Top 10 Revenue generating states:

This analysis displays how the revenue of the company generated from the states of the particular country will be displayed. And we are considering the top 10 states from the country.

Top 10 Revenue...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

+

Stage 1

\$group

1

2

3

4

5

6

7

8

9

10

/**
* _id: The id of the group.
* fieldN: The first field name.
*/
{
 _id: "\$State",
 Revenue: {
 \$sum: "\$ Revenue "
 }
}

Output preview after \$group stage (Sample of 10 documents)

_id: "Val de Marne"
Revenue : 22527

_id: "Washington"
Revenue : 1232427

+

Stage 2

\$sort

1

2

3

4

5

6

/**
* Provide any number of field/order pair
*/
{
 Revenue: -1
}

Output preview after \$sort stage (Sample of 10 documents)

_id: "California"
Revenue : 2490858

_id: "England"
Revenue : 1496497

5. Revenue distribution by Gender:

This output shows the distribution of the content by Gender, displaying the revenue distribution of the company in respect to Male(M) or Female(F).

Revenue Distri...

Generate aggregation

Explain

Run

Options

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Date : "2-19-16"

Year : 2016

Month : "February"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Date : "2-20-16"

Year : 2016

Month : "February"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Date : "2-27-16"

Year : 2016

Month : "February"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Stage1

\$group

1 /**

2 * _id: The id of the group.

3 * fieldN: The first field name.

4 */

5 {

6 _id: "\$Customer Gender",

7 Revenue: {

8 \$sum: "\$ Revenue "

9 },

10 Orders: {

11 \$sum: 1

12 }

13 }

Output preview after \$group stage (Sample of 3 documents)

_id: "M"

Revenue : 4298747

Orders : 17805

_id: "F"

Revenue : 3945069

Orders : 17061

6. Revenue by best performing product categories:

This demonstrates the product category from the Revenue Sales Dataset and show cases the best 10 performing categories along with the revenue generated from the individual categories from the dataset.

Revenue by Be...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

Stage1

\$group

1 /**

2 * _id: The id of the group.

3 * fieldN: The first field name.

4 */

5 {

6 _id: "\$Product Category",

7 Revenue: {

8 \$sum: "\$ Revenue "

9 }

10 }

Output preview after \$group stage (Sample of 4 documents)

_id: null

Revenue : 641.53

_id: "Bikes"

Revenue : 1624558

Stage2

\$sort

1 /**

2 * Provide any number of field/order pair

3 */

4 {

5 Revenue: -1

6 }

Output preview after \$sort stage (Sample of 4 documents)

_id: "Accessories"

Revenue : 5336870

_id: "Bikes"

Revenue : 1624558

7. Profit Analysis by Product Category:

The analysis focuses on the profit generated from each of the product category which shows the importance and weightage of that particular product in the company, in the market.

Profit Analysis ...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Customer Age : 29

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Unit Cost : 3.67

Unit Price : 5

Cost : 11

Revenue : 15

Stage1

\$project

1 /**

2 * specifications: The fields to

3 * include or exclude.

4 */

5 {

6 "Product Category": 1,

7 profit: {

8 \$multiply: [

9 {

10 \$subtract: [

11 "\$ Unit Price ",

12 "\$ Unit Cost "

13]

14 },

15 "\$Quantity"

16]

17 }

18 }

Output preview after \$project stage (Sample of 10 documents)

_id: ObjectId('6a321615dc42fde44e38f409')

Product Category : "Accessories"

profit: 29

_id: ObjectId('6a321615dc42fde44e38f40e')

Product Category : "Clothing"

profit: 8

8. Customer spending Analysis by Age Group:

This demonstrates and displays the spending pattern of the customers present of every kind of age group, and shows the number of orders they have done, along with the total revenue it generated for the company in order to do the analysis.

Customer Spending ...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Customer Gender : "F"

Country : "United States"

State : "Washington"

Product Category : "Accessories"

Cost : 11

Revenue : 15

Stage1

\$bucket

1 /**

2 * groupBy: The expression to group by.

3 * boundaries: An array of the lower bound

4 * default: The bucket name for documents

5 * output: {

6 * outputN: Optional. The output object

7 * }

8 */

9 {

10 groupBy: "\$Customer Age",

11 boundaries: [18, 25, 35, 45, 55, 100],

12 default: "Other",

13 output: {

14 Revenue: {

15 \$sum: "\$ Revenue "

16 },

17 Orders: {

18 \$sum: 1

19 }

20 }

21 }

Output preview after \$bucket stage (Sample of 6 documents)

_id: 18

Revenue : 1138624

Orders : 4800

_id: 25

Revenue : 2851519

Orders : 11606

09. Identifying low performing products:

This analysis output shows the identification of low performing products from the dataset of 2015 and 2016 for the particular company, demonstrating the lowest performing products from using sub category and displays output as total revenue and units sold for the company.

Identifying Lo...

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Date : "2-19-16"
Year : 2016
Month : "February"
Customer Age : 29
Customer Gender : "F"
Country : "United States"
State : "Washington"
Product Category : "Accessories"

Date : "2-20-16"
Year : 2016
Month : "February"
Customer Age : 29
Customer Gender : "F"
Country : "United States"
State : "Washington"
Product Category : "Accessories"

Date : "2-27-16"
Year : 2016
Month : "February"
Customer Age : 29
Customer Gender : "F"
Country : "United States"
State : "Washington"
Product Category : "Accessories"

Stage 1 \$group

1 /**
2 * _id: The id of the group.
3 * fieldN: The first field name.
4 */
5 {
6 _id: "\$Sub Category",
7 TotalRevenue: {
8 \$sum: "\$ Revenue "
9 },
10 UnitsSold: {
11 \$sum: "\$Quantity"
12 }
13 }

Output preview after \$group stage (Sample of 10 documents)

_id: "Fenders"
TotalRevenue : 326113
UnitsSold : 1494

_id: "Bike Stands"
TotalRevenue : 35288
UnitsSold : 304

Identifying Lo...

SAVE

+ CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 2 \$sort

1 /**
2 * Provide any number of field/order pair
3 */
4 {
5 TotalRevenue: 1
6 }

Output preview after \$sort stage (Sample of 10 documents)

_id: null
TotalRevenue : 641.53
UnitsSold : 0

_id: "Bike Racks"
TotalRevenue : 32746
UnitsSold : 204

Stage 3 \$limit

1 /**
2 * Provide the number of documents to lim
3 */
4 10

Output preview after \$limit stage (Sample of 10 documents)

_id: null
TotalRevenue : 641.53
UnitsSold : 0

_id: "Bike Racks"
TotalRevenue : 32746
UnitsSold : 204

10.Best Sales month by each year:

This demonstrates and displays the spending pattern of the customers present of every kind of age group, and shows the number of orders they have done, along with the total revenue it generated for the company in order to do the analysis.

Best Sales Mo...

SAVE

CREATE NEW

EXPORT DATA

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 1\$group

1

/**

2

* _id: The id of the group.

3

* fieldN: The first field name.

4

*/

5

{

6

_id: {

7

year: "\$Year",

8

month: "\$Month"

9

},

10

Revenue: {

11

\$sum: "\$ Revenue "

12

}

13

}

Output preview after \$group stage (Sample of 10 documents)

_id: Object

year: 2015

month: "April"

Revenue : 51592

_id: Object

year: 2015

month: "June"

Revenue : 54110

Stage 2\$sort

1

/**

2

* Provide any number of field/order pair

3

*/

4

{

5

"_id.year": 1,

6

Revenue: -1

7

}

Output preview after \$sort stage (Sample of 10 documents)

_id: Object

Revenue : 641.53

_id: Object

year: 2015

month: "December"

Revenue : 793262

DATA VISUALIZATION USING POWER BI

This is the overall content visualization of all the aggregation analysis we performed in the MongoDB. This also demonstrates clean and clear visuals of the analysis along with particular representation of the charts used for the visualization.

